

IDX EXCHANGE | DATA SCIENCE INTERNSHIP PROGRAM

Automated Valuation Model

Data Science Best Practices

A production-grade standard for building models that generalize to on-market and off-market properties, hold up over time, and reproduce reliably.

Enhanced Edition

Prepared for the Data Scientist Track — Confidential, Internal Use Only

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EXECUTIVE SUMMARY

Twelve Practices Separate a Notebook Model From a Production AVM

A model that scores well on one historical test split is not the same thing as a model that will hold up in production. Production AVMs are re-run every month, valued against properties that never listed, and judged on whether their accuracy is stable — not on a single favorable snapshot. The practices below are organized around that distinction.

Key Takeaways

1. Evaluate the way production actually works: **chronological splits, a held-out validation month, and a rolling-origin backtest across multiple cutoffs — not a single train/test score.**
2. Treat leakage as a structural risk, not a checklist item: ListPrice-family fields, pipeline steps fit only on training data, and CV-safe target encoding all address the same failure mode from different angles.
3. Report **MdAPE** as the headline accuracy metric — it is the most robust to the **right-skewed, multi-scale error distribution that real-estate pricing produces** — alongside R^2 , MAPE, and MAE for a complete picture.
4. Advanced models must beat a Linear Regression baseline by a margin that survives scrutiny, and every result must be reproducible via **fixed seeds, a pinned environment, and a versioned data snapshot.**

BOTTOM LINE

A model is production-ready only when it can be re-run next month, on data it has never seen, and produce metrics that are still defensible.

01 Chronological Splits Are Non-Negotiable — Production AVMs Never See the Future

Use a chronological split, never a random split. Random splits leak future market conditions into training and overstate performance.

- Training set: all months except the most recent complete month.
- Validation set: carve out the second-most-recent complete month from the training set for hyperparameter tuning and model selection — never tune on the test set.
- Test set: the last complete month of available data, touched only once, at the end.

This mirrors how production AVMs work (train on the past, predict the future) and ensures every team is evaluated on the same market conditions.

Rolling-Origin Backtesting

A single train/test cutoff can be lucky or unlucky. Before trusting your metrics, re-run the same pipeline at 2–3 additional historical cutoffs and confirm R^2 , MAPE, and MdAPE stay in a similar range. When a new month's data is released, rerun the full pipeline so the new month becomes the test set — stability over time is itself a metric worth reporting.

BOTTOM LINE

A model that performs well on one cutoff but degrades sharply on the next is overfit to a specific market moment, not learning generalizable pricing structure.

02 Data Quality Starts With Removing Records That Cannot Be True

Remove observations with $\text{ClosePrice} \leq 0$, missing target values, and duplicates. Go further than the obvious cases:

- Logical impossibilities: CloseDate earlier than ListDate, zero or negative square footage, bedroom/bathroom counts inconsistent with living area.
- Duplicate listings for the same property/transaction (same parcel ID or address + close date).
- Log the count and percentage of records removed at each cleaning step — an undocumented 15% drop in rows is a red flag a reviewer should be able to catch.

03 Outlier Thresholds Must Be Learned From Training Data Alone

Compute the 0.5th and 99.5th percentile ClosePrice thresholds using the training set only. Filter the training set using those thresholds and apply the identical, frozen thresholds to the test set. For example: if \$X is the 0.5th percentile of ClosePrice in the training set and \$Y is the 99.5th percentile of ClosePrice in the training set, then \$X and \$Y are the exact cutoffs used to filter the test set as well — the test set's own percentiles are never recomputed or substituted.

- Never recompute percentiles on the test set — that leaks test-set distribution information into your filtering decision.

- Consider a secondary check on **price-per-square-foot outliers**, which catches data-entry errors that a raw price filter misses.

This removes obvious data-entry errors (\$0 sales, accidental extra zeros), rare transactions, and extreme luxury outliers without introducing data leakage.

04 ListPrice and OriginalListPrice Must Be Excluded — They Encode the Answer

These fields are highly correlated with ClosePrice and are only available for on-market properties. The goal is a model that works for **both on-market and off-market homes**, so any feature that only exists because a sale process happened is out of bounds.

Leakage Audit

Before finalizing a feature list, run every candidate feature through this audit:

EXHIBIT 1 FEATURE LEAKAGE AUDIT

Feature	Why It Leaks / Risk	Verdict
ListPrice / OriginalListPrice	Near-perfect correlation with ClosePrice; unavailable for off-market homes	Exclude
Days on Market (DOM)	Only exists once a sale process has unfolded; reflects pricing decisions	Exclude
Price reduction count/flag	Directly encodes seller's pricing behavior relative to ClosePrice	Exclude
Closing/settlement fields	Recorded at or after close; not known pre-sale	Exclude
Tax-assessed value	Fine if assessed independently and lagged; re-assessments can follow a sale	Verify timing
Listing agent / brokerage	Can proxy for pricing strategy rather than intrinsic value	Use cautiously
Square footage, beds/baths, lot size, year built	Physical facts known before any sale	Keep
Latitude/longitude, neighborhood, school zone	Fixed location attributes known before any sale	Keep

Source: IDX Exchange Data Science Practice, internal review standard.

- **Timing check: could this value have been known on the day you're valuing the property, before any sale outcome was known?**
- **Correlation check: a feature with near-1.0 correlation to ClosePrice that isn't an intrinsic characteristic is a leakage suspect, not a gift.**

05 Three Feature Families Drive Most of the Predictive Power

Use intrinsic property characteristics — **square footage, bedrooms, bathrooms, lot size, year built/age, property subtype** — and build outward into **locational and temporal features**.

Locational Features

- Distance to CBD, major employment centers, or top-rated schools computed from latitude/longitude.

- Neighborhood or ZIP-level aggregates (e.g., median price per sqft in the last 12 months), computed on training data only and joined forward to avoid leakage.
- Geohash or spatial clustering as a categorical feature when neighborhood labels are noisy or inconsistent.

Temporal Features

- Month/season encoded with sine/cosine transforms to capture cyclical seasonality without an artificial ordinal jump from December to January.
- Property age at time of sale, not just year built, so the feature reflects condition-relevant aging.

Categorical Encoding

- For high-cardinality categoricals (neighborhood, subtype), avoid naive one-hot encoding blowing up dimensionality; use target encoding computed with cross-validation folds so a category's encoding never sees its own row's label — a common, subtle leakage source.

06 Missingness Is Information — Handle It Deliberately

- Understand the missingness mechanism where possible: missing at random vs. systematically missing (e.g., lot size missing disproportionately for condos).
- Remove columns with excessive missingness when appropriate; document the threshold used and why.
- Impute important features thoughtfully — median/mode imputation is a reasonable default, but consider group-wise imputation (e.g., by property subtype) for better fidelity.
- Add missing-indicator flag features when missingness itself may be informative (e.g., missing lot size often means “condo,” which is itself predictive).
- Never impute the target variable.

07 Leakage Prevention Must Be Structural, Not a Matter of Discipline

All preprocessing — feature engineering, imputation, encoding, scaling, outlier thresholds — must be fit on the training data only and then applied unchanged to the test data. The reliable way to enforce this is structurally:

- Wrap imputers, encoders, and scalers in a scikit-learn Pipeline / ColumnTransformer (or equivalent) so fit only ever sees training data and transform is what touches test data.
- Fit the full pipeline inside each cross-validation fold during model selection, not once on the whole training set — otherwise validation folds leak through shared encodings.
- Treat any groupby/aggregate feature (neighborhood averages, comps-based features) as a fit-on-train transform, identical in spirit to an imputer.

08 MdAPE Is the Right Headline Metric for Skewed, Multi-Scale Price Data

Report R^2 , MAPE, MdAPE, and MAE together — no single metric tells the whole story.

- R^2 : overall variance explained; sensitive to a small number of large-error properties.
- MAPE: average percentage error; intuitive but skewed by outlier percentage errors on very low-priced properties.

- MdAPE: median percentage error; **robust to outliers and often the most representative “typical error” figure.**
- MAE: **dollar-denominated average error**, useful for stakeholders who think in dollars rather than percentages.

Percentage-based metrics are especially appropriate for real estate because dollar errors have different significance at different price levels (\$20K off is nothing on a \$2M home, and everything on a \$150K home).

Why MdAPE Is the Preferred Headline Metric

Real-estate price data typically spans \$150K to \$5M+ and has a right-skewed error distribution (a small number of unusual properties — sparse comps, misclassified subtypes, unreported renovations — produce very large percentage errors). This combination makes MdAPE the most representative single number, for three reasons:

- Scale-invariance: MAE and RMSE are dollar-denominated, so they get dominated by whatever your highest-priced properties are. MAPE and MdAPE normalize by ClosePrice, so a starter home and a luxury home contribute comparably.
- Robustness to outliers: RMSE squares errors before averaging, so a handful of large misses can swamp thousands of good predictions; MAE is linear but still a mean, so it's still pulled by the tail. MdAPE is a median, so a few bad predictions barely move it.
- Skew: because valuation errors are right-skewed, mean-based metrics (MAE, MAPE) read higher than what most properties actually experience. **MdAPE gives a cleaner read on “how the model performs on the median home.”**

RMSE and MAE still earn their place alongside MdAPE: RMSE is the right metric when large errors carry disproportionate cost (e.g., lending risk exposure), and MAE gives a dollar-denominated number that's more intuitive for non-technical stakeholders. Report all four — use MdAPE as the primary figure, and RMSE/MAE to surface tail risk and communicate in dollar terms.

Error Breakdown, Not Just Top-Line Numbers

- Report MAPE/MdAPE separately by **price band (e.g., quintiles) to check whether the model systematically underperforms on luxury or entry-level homes.**
- Report error by geography/neighborhood where sample size allows, to catch localized underperformance that a global metric would hide.

09 Advanced Models Must Beat the Baseline by a Margin That Isn't Noise

Begin with a Simple Linear Regression model as the baseline. Demonstrate that more advanced models (Random Forest, XGBoost, LightGBM, CatBoost, etc.) provide meaningful, statistically noticeable improvements over the baseline — not just a marginal decimal-point gain that could be noise.

- **Tune advanced models using the validation set** (or rolling-origin cross-validation), never the test set.
- **Report baseline and advanced-model metrics side by side in the same table** so the improvement is legible at a glance.

10 Documentation Is What Makes This Work Reproducible By Someone Else

Clearly document data cleaning steps and row counts at each stage, feature engineering logic, feature selection rationale, hyperparameters and how they were chosen, validation strategy, evaluation metrics, known limitations, and concrete next steps for improvement.

- Write this as if a future teammate has to reproduce or extend the work without asking you a single question.
- Call out assumptions explicitly — e.g., “assumed missing lot size implies condo/townhome” — rather than leaving them implicit in code.

11 A Result That Can't Be Reproduced Is an Anecdote, Not a Finding

- Fix random seeds for train/test splitting, cross-validation folds, and model training.
- Pin your environment (requirements.txt or an equivalent lockfile) so package version drift doesn't silently change results.
- Note the data snapshot/version used, so results can be tied back to a specific pull of `rets_property` / `california_sold` rather than “whatever the data looked like that day.”

12 An End-to-End App Is the Real Test of Production-Readiness

Deploy your model with `Streamlit`. Train the model, save the trained model and preprocessing artifacts (model, scaler, encoder, imputer, feature list, preprocessing pipeline), then load them inside `app.py`.

- The app should expose every feature required by the model as an input, transform those inputs exactly as during training, generate predictions, and display the estimated property value.
- Guard against silently mismatched preprocessing: assert the loaded feature list matches the model's expected input schema before predicting.
- This validates that the model is production-ready rather than notebook-only, and it's the fastest way to discover a preprocessing step you forgot to save.

Guiding Principle

Build a model that performs well in production — not just on one historical test split. The strongest AVMs are robust, generalizable, reproducible, well documented, and continue to perform well as new market data becomes available. If you can't re-run your pipeline next month and get consistent metrics, the model isn't done yet.

Self-Check Rubric

Before submitting, confirm each row below is true of your work. This is the bar for a strong submission.

EXHIBIT 2 SUBMISSION SELF-CHECK RUBRIC

#	Practice	Meets Bar
1	Chronological split	Train = all but last complete month; test = last complete month; no shuffling

#	Practice	Meets Bar
2	Rolling-origin backtest	Metrics re-checked on ≥ 2 additional historical month-cutoffs, not just the current one
3	Invalid records removed	ClosePrice > 0, target not null, duplicates and logical impossibilities dropped with a logged count
4	ClosePrice Outlier thresholds	0.5th/99.5th percentile computed on train only, same cutoffs applied to test (e.g., if \$X is the train set's 0.5th percentile and \$Y is its 99.5th percentile, both \$X and \$Y are the cutoffs used to filter the test set as well)
5	No target leakage	ListPrice, OriginalListPrice, DOM, price-reduction flags, and post-close fields excluded; audit documented
6	Feature set	Intrinsic + locational + temporal features; high-cardinality categoricals encoded without leakage
7	Missingness handled	Imputers fit on train only; missing-indicator flags added where informative
8	Leakage-safe pipeline	All fit-time steps inside a Pipeline/ColumnTransformer fit on train only
9	Metrics reported	R^2 , MAPE, MdAPE, and MAE reported; error broken out by price band and/or geography
10	Baseline comparison	Linear Regression baseline reported alongside ≥ 1 gradient-boosted or ensemble model
11	Reproducibility	Random seeds fixed, environment pinned, data snapshot version noted
12	Documentation	Assumptions, limitations, and next steps written down, not just code comments

Source: IDX Exchange Data Science Practice, internal review standard.